Data Quality Plan (DQP)

Dataset: geocoded_train.csv → train-clean-final.csv
Module: COMP47350 Group Project - Part (2) DQP
Scope: Feature-level issue resolution, chosen cleaning actions, justification, final feature typing, and cleaned CSV export.

1. Executive Summary

This report defines the Data Quality Plan (DQP) implemented in Part 2 of project.ipynb. It transforms the geocoded training dataset into a modelling-ready cleaned CSV through explicit, auditable, and justified preprocessing decisions.

The input to DQP is geocoded_train.csv with 54,000 rows and 14 columns. The five extra columns are geocoding artifacts used to derive better location information for Dublin properties. The final exported output is train-clean-final.csv with 53,991 rows, 10 columns, no missing values, and a fixed analytical schema for downstream modelling.

Main risks addressed: duplicate rows, unusable raw address text, high missingness in Eircode and Property Size Description, VAT-related price inconsistency, and severe price outliers.

Main policy choice: preserve interpretable business signals while replacing fragile raw fields with compact engineered variables.

Main output: a complete cleaned training set with finalized feature types and a robust target variable Price_Log.

2. DQP Scope and Baseline Input

2.1 Input Dataset for Cleaning

The notebook starts the executable DQP stage from geocoded_train.csv rather than directly from the raw RPPR file because address geocoding was performed offline. This separates a slow external-API step from the deterministic cleaning steps implemented and logged in the notebook.

At load time, the dataset contains 54,000 rows × 14 columns: the original 9 RPPR variables plus geocode_status, latitude, longitude, geocode_provider, and dublin_district. Duplicate rows equal 9. The same two major missingness issues from DQR remain present: Property Size Description is 94.75% missing and Eircode is 68.72% missing.

Baseline Item	Observed Value
Rows × columns	54,000 × 14
Duplicate rows	9
Property Size Description missing	51,164 (94.75%)
Eircode missing	37,108 (68.72%)
Geocoding columns	Only populated for Dublin rows by design

Purpose of baseline	Before-clean reference for the cleaning log

2.2 Cleaning Objectives

List every input feature, its type, its data quality issue, and the chosen resolution.
Make final decisions on analytical feature types for all retained variables.
Ensure no NaN values remain in the exported cleaned dataset.
Save the final cleaned CSV with a self-explanatory filename.

3. Feature-Level Decisions

Table 3.1 summarizes the final decision taken for every feature present at the start of the DQP stage. Detailed justification follows immediately after the table so the document remains readable while still satisfying the per-feature requirement in the specification.

Feature	Input Type	Final Decision	Final / Derived Type
Date of Sale (dd/mm/yyyy)	string/object	Parse, derive Sale Year and Sale Month, then drop raw date	Sale Year: Int64; Sale Month: Int64
Address	string/object	Extract is_apartment and use offline geocoding to support location; then drop raw field	is_apartment: int64
County	string/object	Merge with dublin_district to build location; then drop source field	location: category
Eircode	string/object	Drop due to high missingness and redundancy after location engineering	Dropped
Price (€)	string/object	Parse to numeric; preserve for audit; derive Price_Log target through VAT adjustment + clamping + log	Price (€): float64; Price_Log: float64
Not Full Market Price	string/object	Retain as categorical predictor	category
VAT Exclusive	string/object	Retain as categorical predictor and use in VAT-adjustment rule	category
Description of	string/object	Retain as categorical predictor and use in	category

Property		VAT-adjustment rule	
Property Size Description	string/object	Drop due to 94.75% missingness	Dropped
geocode_status	string/object	Drop after QA	Dropped
latitude	float/object	Drop after district derivation	Dropped
longitude	float/object	Drop after district derivation	Dropped
geocode_provider	string/object	Drop after QA	Dropped
dublin_district	string/object	Use to construct location, then drop source field	location: category

3.1 Date of Sale (dd/mm/yyyy)

Issue: stored as raw text and unusable for modelling in its original form. Options considered: keep as string, parse to datetime only, or derive compact temporal variables. Chosen solution: parse with dayfirst=True, extract Sale Year and Sale Month, and drop the original date column. This retains the useful time signal while avoiding redundancy.

3.2 Address

Issue: near-unique free text with extreme cardinality. Options considered: one-hot encode raw text, drop immediately, or engineer compact features. Chosen solution: extract is_apartment from keywords and use offline geocoding to support Dublin district derivation. The raw field is then dropped because the engineered signals are more stable and interpretable.

3.3 County and dublin_district

Issue: County is too coarse for Dublin pricing, while dublin_district is only meaningful for Dublin rows. Options considered: keep County only, keep both fields separately, or merge them into a unified location feature. Chosen solution: create location so Dublin rows become Dublin_<district> and all other rows retain their County name. This gives one interpretable geography variable with 51 categories.

3.4 Eircode

Issue: 68.72% missing and high cardinality. Options considered: impute as text, use prefix-only encoding, or drop. Chosen solution: drop the field once location has been engineered because the remaining postcode signal is too sparse and unstable relative to its complexity.

3.5 Price (€)

Issue: currency-formatted text, heavy right skew, and extreme outliers. Options considered: use raw numeric price, log-transform raw price only, or create a standardised robust target. Chosen solution: parse to float, preserve the original declared price for auditability, create VAT-adjusted price where needed, clamp the adjusted price using training-only percentile bounds, and model log1p of the clamped value as Price_Log.

3.6 Not Full Market Price

Issue: binary text field not yet typed. Options considered: drop, map to numeric, or retain as category. Chosen solution: retain as a categorical predictor because it is complete, low-cardinality, and likely informative.

3.7 VAT Exclusive and Description of Property

Issue: together they reveal a price-definition inconsistency for new-build properties reported VAT-exclusive. Options considered: ignore the inconsistency, drop one of the variables, or apply a rule-based correction. Chosen solution: retain both as categorical predictors and use them jointly to identify new dwellings requiring a 13.5% VAT uplift. This preserves business meaning and standardises the price scale.

3.8 Property Size Description

Issue: 94.75% missing. Options considered: fill with Unknown, impute a dominant category, use missingness as a signal, or drop. Chosen solution: drop entirely because the observed coverage is too low for reliable imputation or trustworthy modelling value.

3.9 Geocoding auxiliary fields: geocode_status, latitude, longitude, geocode_provider

Issue: these columns describe the preprocessing workflow rather than the property itself; they are also blank for most non-Dublin rows by design. Options considered: keep for audit only, use them directly as predictors, or drop after QA. Chosen solution: drop them after validation because they are intermediate artifacts, not stable business features.

4. Implemented Cleaning Workflow and Outcomes

4.1 Duplicate removal

9 exact duplicate rows are removed first, reducing the dataset from 54,000 to 53,991 rows. This is done before dropping Address so that deduplication reflects the full original record.

4.2 is_apartment extraction

A binary feature is derived from Address using apartment/apt/unit/flat keywords. 3,090 records (5.7%) are flagged as apartment-style dwellings.

4.3 Dropping geocoding auxiliaries and raw Address

geocode_status, latitude, longitude, geocode_provider, and Address are removed after their role is complete.

4.4 Basic standardisation

Whitespace is normalised; Date of Sale is parsed; Price (€) is converted to float; Sale Year and Sale Month are created; low-cardinality fields are cast to category.

4.5 location construction

County and dublin_district are merged into location. The resulting feature contains 51 unique categories and replaces both source fields.

4.6 Core field validation

Rows are checked for positive price, valid parsed date, and non-empty location. No rows fail these rules, so 0 rows are rejected.

4.7 VAT adjustment

For new dwellings with VAT Exclusive = Yes, Price_Adjusted_VAT = Price × 1.135. This affects 9,513 records.

4.8 Price clamping and log transform

Training-only cut points are computed on Price_Adjusted_VAT: p05 = €60,000 and p95 = €721,137.50. 5,333 records are clamped. Price_Log = log1p(Price_Final) becomes the final modelling target.

4.9 Dropping high-missingness and redundant fields

Property Size Description, Eircode, the raw date column, Price_Adjusted_VAT, and Price_Final are removed once their information has been transferred or judged unusable.

4.10 Final missing-value policy and export

A final policy of categorical → Unknown and numeric → median is defined, but no remaining missing values are found in practice. The cleaned dataset is exported as train-clean-final.csv.

Step	Rows After	Columns After	Outcome
Load baseline	54,000	14	geocoded_train.csv loaded
Drop duplicates	53,991	14	9 rows removed
Extract is_apartment	53,991	15	binary dwelling-type signal created
Drop Address + auxiliaries	53,991	10	intermediate columns removed
Basic standardisation	53,991	12	typed fields and temporal features created
Construct location	53,991	11	51-category geography feature created
Core validation	53,991	11	0 rows rejected
VAT adjustment	53,991	13	9,513 rows adjusted
Price clamping + log target	53,991	15	Price_Log created
Drop redundant columns	53,991	10	final compact schema

Final missing-value check	53,991	10	0 NaN values remain

5. Final Cleaned Dataset Specification

The exported DQP output contains one audit column, eight predictor columns, and one modelling target column. All feature types are fixed and no missing values remain.

Retained Column	Final Type	Role	Reason for Retention
Price (€)	float64	Audit column	Preserves the originally declared price for back-checking and transparency.
Not Full Market Price	category	Predictor	Complete, low-cardinality business indicator.
VAT Exclusive	category	Predictor	Retained for interpretation alongside VAT standardisation.
Description of Property	category	Predictor	Captures dwelling class and supports the VAT rule.
is_apartment	int64	Predictor	Compact property-type signal derived from Address.
Sale Year	Int64	Predictor	Captures long-run market trend.
Sale Month	Int64	Predictor	Captures within-year timing.
location	category	Predictor	Unified geography variable with Dublin district granularity.
vat_adjusted_flag	int64	Predictor	Marks which records were VAT-standardised.
Price_Log	float64	Target	Final target: log1p of VAT-adjusted clamped price.

5.1 Final Validation Outcomes

Final shape: 53,991 rows × 10 columns.
Total remaining missing values: 0.
All Price (€) values are positive.
All Price_Log values are positive.
Saved file: train-clean-final.csv.

6. Conclusion

This DQP satisfies the project requirement to identify feature-level issues, consider alternative solutions, justify the chosen actions, apply those actions, finalise feature types, and export a cleaned CSV with no missing values. The resulting dataset is compact, interpretable, and aligned with the modelling needs of the project.

The most important design choices are to replace fragile raw location text with compact engineered geography, standardise VAT-exclusive new-build prices before modelling, and control extreme target values through training-only clamping. Together these actions materially improve consistency, robustness, and reproducibility.

Appendix A: Alignment with Part 2 of project.ipynb

§2.1: offline geocoding workflow documented and used as source of dublin_district.
§2.2: baseline check records shape, dtypes, missingness, and duplicate count.
§2.3: duplicate removal deletes 9 rows.
§2.4–§2.4.1: is_apartment extracted before Address and geocoding auxiliaries are dropped.
§2.5: standardisation performs whitespace cleaning, date/price parsing, temporal extraction, and category casting.
§2.6: County and dublin_district are merged into location.
§2.7: core validation rejects 0 rows.
§2.8: VAT adjustment affects 9,513 rows at rate 13.5%.
§2.9–§2.9.1: p05 = €60,000 and p95 = €721,137.50 are used for clamping; Price_Log is created.
§2.10–§2.13: sparse/redundant fields are dropped, final validation passes, and train-clean-final.csv is exported.